

TRUNG TUAN MAI

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EDUCATION

Bachelor of Science, Software Engineering Schulich School of Engineering, University of Calgary	Expected April 2027
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CORE SKILLS

Programming: Python, Java, JavaScript, C/C++
Backend: Flask, Spring Boot, REST APIs
Databases: MongoDB, MySQL, H2
Tools: Docker, Git, Gradle, Linux
Concepts: API design, event-driven systems, data pipelines, system debugging

PROJECTS

Data Ingestion & Backend System (Web-Based Systems Course)	February 2026
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- Engineered a Flask backend with REST APIs to ingest and normalize multi-source datasets into MongoDB, enabling consistent data storage across heterogeneous inputs.
- Implemented schema validation and unique-key deduplication to enforce data consistency and prevent record conflicts across sources.
- Designed an ETL-style pipeline combining scraping, parsing, transformation, and persistence to automate data ingestion workflows and reduce manual processing.
- Diagnosed and resolved API-database integration issues to improve system reliability under concurrent service execution.

Backend Job-Market Simulation System (Software Architecture Course)	February 2026
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- Engineered a Flask and SQLAlchemy backend managing player progression and application state, enabling persistent tracking of user lifecycle data.
- Designed REST APIs for lifecycle management, activity tracking, and application history retrieval, supporting reliable client-server state synchronization.
- Implemented rule-based hiring logic enforcing progression constraints and deterministic outcome computation across system states.

Movie Ticket Reservation System (Principle of Software Design Course)	April 2025
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- Developed a Java-based reservation system using JDBC and MySQL to manage bookings, seats, and transactions, enabling consistent end-to-end reservation workflows.
- Implemented booking and cancellation workflows, ensuring synchronized updates between application logic and database state.
- Optimized relational queries to improve transaction consistency and reduce data synchronization issues.

EXTRA-CURRICULARS

Financial Transaction Processing Backend JP Morgan Chase Virtual Experience in Forage	January 2026
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- Developed Spring Boot backend services exposing REST endpoints for account balance retrieval and persistence, enabling consistent financial state access.
- Integrated Kafka-based message listeners to simulate asynchronous transaction processing, improving system scalability and decoupling service interactions.
- Structured repository-based data access using Spring Data JPA to ensure maintainable and consistent database interaction.

WORK EXPERIENCE

Server - Pho Pham Restaurant	Present
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- Coordinated high-volume operations handling 100+ transactions per shift while maintaining workflow efficiency and accuracy.
- Resolved real-time operational issues in a fast-paced environment through structured problem-solving and team coordination.